WRITTEN ASSIGNMENT

CSEBCSCSAS01 - Computer Science and Society

Source: Written assignment task sheet, CSEBCSCSAS01

1. Understanding the assignment

2. Choosing and narrowing a topic

3. Building a research question

4. Finding and using sources

5. Argument, structure, and ethics

6. Topic clinics: newsfeeds, energy, drones

7. Academic integrity and AI use

8. Outline workshop and Q&A

Learning outcomes

- Explain what the written assignment asks students to do
- Choose a feasible topic and formulate a focused research question
- Distinguish description, analysis, critical discussion, and evaluation
- Develop a defensible paper structure with sources and argument logic
- Apply topic-specific thinking to algorithms, critical infrastructure, drones, or an own research question
- Avoid plagiarism and use AI tools responsibly within academic expectations

Opening reflection

ACTIVITY

What makes a written assignment difficult?

1. Finding sources	2. Narrowing the topic	3. Critical discussion	4. Writing academically

10 minutes

PART 1

Understanding the examination task

From task sheet to writing strategy

What the task sheet actually requires

- Students choose one of four topics for the written assignment
- The course book is the starting point, but additional sources are required
- The assignment is not a summary task; it asks for explanation, critical discussion, and independent research
- Evaluation should follow the writing guidelines and formal academic criteria

Choose one topic
+ course book
+ additional sources
+ independent argument

Source: Task sheet: Sections 1 and 2

The four available paths

- Task 1: Newsfeeds and Facebook algorithms
- Task 2: Critical infrastructure in the energy sector
- Task 3: Dual use of drone technology
- Task 4: Own research question approved in advance
- All four require a connection to Computer Science and Society

Newsfeeds

Energy
critical infrastructure

Drones
dual use

Own research
question

One paper - one clear argument

Source: Task sheet: Sections 1.1-1.4

The task verbs matter

- Explain: clarify concepts, mechanisms, and context
- Describe: present concrete risks, actors, regulations, or measures
- Critically discuss: evaluate assumptions, consequences, trade-offs, and ethical tensions
- Develop: create a justified research question, outline, and procedure for Task 4

Explain

Describe

Critically discuss

Develop

Task decoding exercise

ACTIVITY

Take one task and mark the verbs, objects, and expected evidence.

1. Read task
2. Underline verbs
3. List outputs
4. Identify sources

12 minutes

PART 2

Choosing a topic and narrowing the scope

From broad field to manageable paper

A good topic is not the same as a good paper

- A topic is broad: “drones”, “Facebook”, “critical infrastructure”
- A paper needs a focus: country, sector, technology, ethical issue, regulation, actor group
- A feasible paper asks one main question and answers it with evidence
- The scope should match the word limit, time available, and source availability

Broad topic → focused question → evidence-based answer

Narrowing dimensions

- Geography: one country, EU context, or a justified comparison
- Actor: platform operator, user, regulator, energy provider, drone operator, citizen
- Technology: recommender algorithm, smart grid, SCADA, UAV sensors, autonomous navigation
- Societal issue: privacy, safety, fairness, security, accountability, democracy, dual use
- Time: current situation, future threats, historical development, or regulatory transition

Where?

Who?

Which technology?

Which societal issue?

Which time frame?

Examples of weak vs stronger focus

- Weak: “Facebook algorithms are bad for society.”
- Stronger: “How do engagement-based ranking signals in Facebook Feed create ethical tensions between personalization and democratic information quality?”

- Weak: “Cybersecurity in energy is important.”
- Stronger: “Which IT security risks are most relevant for smart-grid operators in Germany, and how do current countermeasures address ransomware and supply-chain threats?”

A strong focus contains: technology + context + issue + evaluative angle

Scope ladder

ACTIVITY

Make your topic narrower in four steps.

1. Broad topic

2. Add context

3. Add ethical/security angle

4. Turn into question

15 minutes

Feasibility checklist

- Can I define all key concepts clearly?
- Can I find at least 6-10 credible sources beyond the course book?
- Can I answer the question without needing confidential data?
- Can I show both technical and societal understanding?
- Can I write a balanced critical discussion rather than a one-sided opinion?

If any answer is no:
reduce scope or change angle

Learning Control Question

Which is usually the best scope for a term paper?

a) The whole history of AI

b) One clearly defined problem with evidence

c) All ethical issues in technology

d) A list of interesting facts

Correct answer: b

PART 3

Research questions and thesis statements

How to guide the paper

Research question vs title vs thesis

- Title: signals the topic and focus
- Research question: defines what the paper investigates
- Thesis statement: gives the central answer or position that the paper defends
- The introduction should make all three visible

Title
What is this about?

Research question
What do I investigate?

Thesis
What do I argue?

Useful research-question patterns

- How does X technology affect Y social value in Z context?
- What risks arise when X is used in Y sector, and how effective are current safeguards?
- To what extent does regulation X address problem Y?
- Which ethical trade-offs emerge between A and B in the use of technology X?

Good questions are answerable, analytical, and bounded

From research question to argument

- Question: “How does Facebook Feed ranking reflect competing values?”
- Possible answer: ranking optimizes relevance and engagement but can conflict with transparency, autonomy, and information quality
- Body sections then prove this answer through mechanisms, examples, literature, and ethical analysis
- The conclusion returns directly to the research question

Every body section should help answer the research question

Research-question clinic

ACTIVITY

Draft one research question and improve it with peer feedback.

1. Draft
2. Check scope
3. Add concept
4. Revise

20 minutes

Common problems in research questions

- Too broad: cannot be answered in one assignment
- Too descriptive: only asks “what is” without evaluation or analysis
- Too normative without evidence: “Should all drones be banned?”
- Too technical without society: “How does a CNN detect objects?”
- Too personal: depends mainly on opinion rather than literature and evidence

Fix by adding: context, criteria, evidence, and a clear evaluative angle

Learning Control Question

What does a thesis statement do?

a) Lists all references

b) States the central answer or position

c) Replaces the conclusion

d) Avoids critical discussion

Correct answer: b

PART 4

Finding, reading, and using sources

From web search to academic evidence

Source types students should combine

- Course book: conceptual foundation
- Peer-reviewed articles: academic argument and theory
- Official sources: regulations, agencies, standards, and public reports
- Technical documentation: how systems are described by providers
- Credible journalism: current cases and public debate, used carefully

Course book

Academic literature

Official reports

Technical documents

Journalism

Evidence quality ladder

- Strong: peer-reviewed research, official regulation, standards, regulator reports
- Useful but contextual: company transparency pages, technical blogs, white papers
- Use with caution: media articles, opinion pieces, advocacy reports
- Weak: unsourced blogs, random summaries, AI-generated claims without verification
- Best papers triangulate claims across different source types

Do not rely on one source type only

Search strategy: from keywords to concepts

- Start with the assignment vocabulary: newsfeed, algorithmic ranking, critical infrastructure, energy sector, dual use, drones
- Add academic concepts: ethics, accountability, transparency, privacy, safety, security, fairness
- Add context: EU, Germany, your country, Facebook, energy grid, UAV regulation
- Use citation chasing: one good paper leads to more relevant sources

Keyword set = technology + context + societal issue

Reading sources efficiently

- First read abstract, introduction, conclusion, and headings
- Ask: What is the claim? What evidence supports it? What method is used?
- Extract definitions, mechanisms, arguments, and limitations
- Write notes in your own words immediately to avoid accidental plagiarism
- Record citation details while reading, not at the end

Read for argument, not only for quotations

Source evaluation exercise

ACTIVITY

Compare two sources for credibility and usefulness.

1. Identify author
2. Identify evidence
3. Check date/context
4. Decide use

15 minutes

Integrating sources into your writing

- Do not drop citations at the end of paragraphs without explaining relevance
- Use sources to define concepts, support claims, compare positions, and show evidence
- After a citation, add your own interpretation: why does this matter for your question?
- Balance direct quotations with paraphrase; most academic writing should be your own synthesis

Source + interpretation = academic argument

Learning Control Question

Which source combination is strongest?

a) Only Wikipedia

b) Course book + peer-reviewed articles + official sources

c) Only company marketing pages

d) Only generative AI output

Correct answer: b

PART 5

Structure and argumentation

How to make the paper coherent

A robust assignment structure

- 1. Introduction: problem, relevance, research question, structure
- 2. Background: concepts, technology, societal context
- 3. Main analysis: mechanisms, risks, actors, evidence
- 4. Critical discussion: ethical trade-offs, limitations, counterarguments
- 5. Conclusion: answer the question, implications, future outlook

Introduction
→ Background
→ Analysis
→ Critical discussion
→ Conclusion

What belongs in the introduction?

- Introduce the problem and why it matters for Computer Science and Society
- Define the specific scope: technology, sector, country, or actor group
- State the research question clearly
- Preview the structure of the paper
- Avoid long history sections unless history is part of your question

The introduction is a contract with the reader

What is critical discussion?

- Not just negative criticism
- It weighs benefits, risks, values, uncertainties, and stakeholder perspectives
- It asks whether evidence supports a claim and what assumptions remain
- It considers counterarguments and limitations
- It connects technical design choices to societal consequences

Critical = balanced, evidence-based, and reflective

Argument logic: claim, evidence, reasoning

- Claim: the point you want the reader to accept
- Evidence: literature, examples, data, technical documentation, regulation
- Reasoning: explanation of how the evidence supports the claim
- Counterpoint: an alternative interpretation or limitation
- Mini-conclusion: link back to the research question

Claim

Evidence

Reasoning

Counterpoint

Mini-conclusion

Paragraph repair

ACTIVITY

Turn a descriptive paragraph into an analytical paragraph.

1. Find claim
2. Add evidence
3. Add reasoning
4. Add link back

15 minutes

Common structure mistakes

- The paper becomes a list of facts rather than an argument
- The research question appears only in the introduction and is forgotten later
- Technical explanation is too long and ethical analysis is too short
- The conclusion introduces new evidence instead of synthesising the answer
- Sections do not have clear transitions

Every section should answer: why is this here?

Learning Control Question

What is the best function of the conclusion?

a) Introduce a new topic

b) Answer the research question based on the analysis

c) Repeat the introduction word for word

d) Add uncited facts

Correct answer: b

PART 6

Topic 1: Newsfeeds

Algorithms, values, and ethics

Task 1: What students need to cover

- Explain how Facebook uses algorithms to generate the newsfeed
- Use examples to discuss values or goals reflected in the algorithms
- Critically discuss ethical implications
- Useful angles: personalization, engagement, relevance, ranking signals, transparency, autonomy, misinformation

Mechanism
+ values
+ ethics

Source: Task sheet: Section 1.1

Current context: feed ranking and recommender systems

- Social media feeds are not neutral chronological lists; they are ranked and personalized information environments
- Meta describes Facebook Feed as an AI system that predicts what users may find valuable and relevant
- The DSA requires transparency around recommender systems for online platforms
- A strong paper links technical ranking choices to societal information quality

Ranking systems organize attention

Source: Meta Transparency Center, 2025; European Commission DSA

Possible values embedded in newsfeed algorithms

- Relevance: show content users are predicted to value
- Engagement: encourage interaction, time spent, reactions, comments, sharing
- Safety: downrank or remove harmful or policy-violating content
- Commercial value: advertising, creator economy, platform retention
- User control: options to influence or modify feed ranking

Relevance

Engagement

Safety

Commercial value

User control

Ethical issues for Task 1

- Autonomy: are users choosing information, or being steered?
- Transparency: can users understand why they see content?
- Fairness: whose voices are amplified or suppressed?
- Democracy: can ranking shape public debate and polarization?
- Responsibility: who is accountable for harmful outcomes?

Ethics = values in conflict

Newsfeed case discussion

ACTIVITY

Should a platform optimize for engagement, user satisfaction, or democratic information quality?

1. Stakeholders

2. Benefits

3. Risks

4. Your position

20 minutes

PART 7

Topic 2: Critical infrastructure in energy

Security, resilience, and societal dependency

Task 2: What students need to cover

- Explain critical infrastructure in the energy sector
- Describe specific IT security risks for the energy sector in your country
- Discuss measures used to counter these risks
- Identify relevant players and future threat scenarios

Definition
+ country risks
+ measures
+ actors
+ future threats

Source: Task sheet: Section 1.2

Why energy infrastructure is a Computer Science and Society topic

- Energy systems depend on digital monitoring, control, communication, and automation
- Failures can affect households, hospitals, transport, water, communication, and public safety
- Cybersecurity is therefore not only a technical problem; it is a social resilience problem
- Students should connect IT risk to societal consequences

Digital dependency → systemic risk

Typical IT security risks in the energy sector

- Ransomware and operational disruption
- Supply-chain compromise affecting vendors and software updates
- Remote-access weaknesses and poor segmentation
- Attacks on industrial control systems and operational technology
- Insider threats, human error, and misconfiguration
- Disinformation or coordinated attacks during crisis situations

Think IT + OT + people + supply chain

Source: ENISA Threat Landscape 2024

Relevant players to identify

- Energy producers, grid operators, distribution system operators, and suppliers
- National cybersecurity agencies and critical-infrastructure regulators
- EU-level actors, including ENISA and regulatory frameworks such as NIS2
- Technology vendors, cloud providers, equipment manufacturers, and incident-response teams
- Citizens and dependent sectors as affected stakeholders

Operators

Regulators

EU agencies

Vendors

Citizens

Future threat scenarios

- More complex attacks against smart grids and distributed energy resources
- AI-assisted phishing, reconnaissance, malware development, and vulnerability discovery
- Cascading failures across energy, telecoms, cloud, transport, and public administration
- Geopolitical conflict and hybrid threats against critical services
- Climate-related stress combined with cyber incidents

Future threats are combined threats

Energy-sector threat map

ACTIVITY

Map one energy-system threat from attacker to societal impact.

1. Asset	2. Threat	3. Vulnerability	4. Consequence

20 minutes

PART 8

Topic 3: Dual use of drones

Civil innovation, security, and ethics

Task 3: What students need to cover

- Explain drones as remotely controlled unmanned aerial vehicles
- Clarify why drones are considered dual-use assets
- Describe suitable purposes and applications
- Describe regulations in your country
- Explore ethical challenges arising from drone use

Technology
+ dual use
+ regulation
+ ethics

Source: Task sheet: Section 1.3

What “dual use” means

- A dual-use technology can serve beneficial civilian purposes and harmful or military/security purposes
- Drones can support logistics, agriculture, environmental monitoring, disaster response, journalism, and inspection
- The same capabilities can support surveillance, targeting, smuggling, harassment, or weaponization
- The ethical issue is not only the device, but the context, payload, autonomy, data, and purpose

Same technology
Different purposes
Different risks

Regulatory angle for drones

- Students should identify the relevant national or EU rules for drone operation
- In the EU, EASA distinguishes categories such as open, specific, and certified operations based on risk
- Regulation may include operator registration, pilot competence, operational limits, insurance, privacy, and restricted zones
- A good paper explains not only the rule, but which risk the rule tries to manage

Regulation follows operational risk

Source: EASA civil drone guidance

Ethical issues for drone use

- Privacy: aerial data collection can be difficult for affected persons to notice or avoid
- Safety: crashes, airspace conflicts, and failures can harm people or property
- Accountability: responsibility may involve operator, manufacturer, software provider, and regulator
- Military/security use: distance, surveillance, targeting, and escalation raise serious moral questions
- Equity: who benefits from drone deployment and who bears the risk?

Ethics depends on use, context, control, and harm

Dual-use matrix

ACTIVITY

Place drone applications on a benefit-risk matrix.

1. Civil benefit
2. Potential harm
3. Regulation
4. Ethical judgement

20 minutes

PART 9

Task 4: Own research question

Freedom with responsibility

Task 4: What students need to do

- Develop a research question in the subject area of the module
- Make a clear reference to the course script
- Introduce a topic from practice or a personal professional context if relevant
- Discuss and get approval for the research question, rough outline, and procedure in advance

Own question ≠ anything goes

Source: Task sheet: Section 1.4

Good Task 4 topics

- Fit Computer Science and Society, not only pure technical implementation
- Have a clear social, ethical, economic, legal, or political dimension
- Can be answered with credible sources
- Are narrow enough for the assignment
- Avoid confidential organisational data unless anonymised and permitted

Module fit + feasibility + approval

Possible current Task 4 directions

- Generative AI in education and assessment integrity
- AI agents and responsibility in automated workflows
- Digital identity, biometric systems, and civil rights
- Algorithmic management in platform work
- Digital divide and access to AI-enabled services
- Data centres, AI energy demand, and sustainability

Current topic must still be tied to course concepts

Task 4 pitch

ACTIVITY

Prepare a 90-second pitch for an own research question.

1. Topic	2. Course link	3. Question	4. Sources

18 minutes

Learning Control Question

For Task 4, what must be approved in advance?

a) Only the final bibliography

b) Research question, rough outline, and procedure

c) Only the title font

d) Nothing

Correct answer: b

PART 10

Ethical analysis frameworks

How to make the discussion deeper

Stakeholder analysis

- Identify who designs, deploys, regulates, uses, and is affected by the system
- Separate direct users from affected non-users
- Ask who benefits, who bears risk, and who has voice or control
- Use stakeholder analysis to avoid abstract or one-sided ethical discussion

Designers
Operators
Users
Affected persons
Regulators

Value trade-off analysis

- Common values: safety, privacy, autonomy, fairness, efficiency, innovation, security, accountability
- Technology often improves one value while weakening another
- A critical paper names the trade-off and explains why it matters
- The strongest papers define criteria for judgement rather than only stating opinions

Example: personalization vs autonomy
security vs privacy
innovation vs safety

Responsibility analysis

- Who has causal responsibility for harm?
- Who has role responsibility because of their position?
- Who has legal responsibility under regulation?
- Who has moral responsibility to prevent foreseeable harm?
- How can responsibility be documented, audited, or challenged?

Responsibility is often distributed

Risk-benefit analysis

- Benefits: efficiency, access, safety, innovation, cost reduction, new capabilities
- Risks: harm, misuse, bias, opacity, surveillance, dependency, security failure
- Assess likelihood, severity, reversibility, and affected groups
- Discuss safeguards and residual risks, not only risks in isolation

Risk without mitigation is incomplete analysis

Ethics framework application

ACTIVITY

Apply stakeholder + value trade-off analysis to your chosen topic.

1. Stakeholders	2. Values	3. Conflict	4. Judgement

18 minutes

Learning Control Question

Which question best supports ethical analysis?

a) What does the technology cost?

b) Who benefits and who is exposed to risk?

c) Which font should I use?

d) Can I cite one source only?

Correct answer: b

PART 11

Academic integrity, plagiarism, and AI tools

Independent work in a generative AI era

The task sheet explicitly warns about plagiarism

- Examination tasks are copyrighted and should not be published on third-party platforms
- Submitted assignments are checked using plagiarism software
- Students should not share solutions because this can create suspicion of plagiarism
- Independent preparation is part of the examination performance

Do not share solutions
Do not submit copied text
Document your sources

Source: Task sheet: copyright and plagiarism note; Section 3

What counts as plagiarism?

- Copying text without quotation marks and citation
- Paraphrasing too closely while keeping the original structure or wording
- Using ideas, data, figures, or arguments without attribution
- Reusing another student’s work or shared online solution
- Submitting AI-generated content as if it were your own independent work when not allowed or not disclosed

Plagiarism is about words, ideas, and authorship

Responsible AI use for writing support

- Use AI tools for brainstorming, language improvement, or self-testing only if allowed by course/exam rules
- Never trust AI-generated facts, references, quotations, or legal claims without verification
- Do not outsource your analysis, argument, or final text to an AI tool
- Keep records of how tools were used if disclosure is required
- Your submitted work must represent your own understanding and reasoning

AI can assist process; it must not replace authorship

Citation and paraphrase practice

- Read the source, close it, then write the idea in your own words
- Compare with the original to ensure your phrasing and structure are genuinely different
- Cite the source even when paraphrasing
- Use direct quotation sparingly and only when exact wording matters
- Connect each source to your own argument

Paraphrase ≠ synonym replacement

Learning Control Question

Which practice is safest academically?

a) Use AI-generated references without checking

b) Copy a shared solution

c) Write from sources with clear citations and your own argument

d) Submit a draft written by a friend

Correct answer: c

PART 12

Workshop: building the outline

From plan to submission-ready paper

Outline template for students

- Working title
- Research question
- Thesis statement or expected answer
- 3-5 main sections with purpose of each section
- Initial source list with source types
- Risks: scope problems, missing sources, unclear ethics angle

A good outline is an argument map

Example outline: newsfeeds

- Introduction: why algorithmic feeds matter for information society
- Background: recommender systems and Facebook Feed ranking
- Analysis: values reflected in ranking and personalization
- Ethical discussion: autonomy, transparency, public discourse, responsibility
- Conclusion: answer the research question and identify limitations

Do not turn this into a product description; keep the ethical argument central

Example outline: energy infrastructure

- Introduction: digital dependency and national energy resilience
- Background: critical infrastructure and energy-sector IT/OT systems
- Risks: cyber threats and sector-specific vulnerabilities in one country
- Measures and actors: regulation, operators, agencies, technical safeguards
- Future scenarios and conclusion

Link every technical risk to societal impact

Example outline: drones

- Introduction: why drones are a dual-use technology
- Background: UAV basics and application areas
- Regulation: relevant national/EU framework and operational categories
- Ethical discussion: privacy, safety, accountability, military/security use
- Conclusion: balanced judgement and future governance needs

Dual use requires balanced analysis of benefit and harm

Outline workshop

ACTIVITY

Build a one-page outline for your chosen task.

1. Question

2. Thesis

3. Sections

4. Sources

30 minutes

Peer review

ACTIVITY

Exchange outlines and test them against the assignment criteria.

1. Clarity

2. Scope

3. Evidence

4. Critical angle

20 minutes

Final submission checklist

- I answer exactly one selected task or approved own question
- My research question is visible and focused
- My structure follows a clear argument, not a list of facts
- I use course concepts and additional credible sources
- I critically discuss ethical, social, security, or regulatory implications
- All borrowed words, ideas, data, and figures are cited

Before submission: check task fit, sources, argument, citations

Closing reflection

- Which task are you most likely to choose?
- What is your current research question?
- Which source type do you still need?
- What is the strongest ethical or societal tension in your topic?
- What is your next concrete writing step?

Write one next action before leaving

Learning Control Question

What should students leave this lecture with?

a) A complete final paper

b) A clear task choice, draft research question, and outline direction

c) Permission to ignore citations

d) A copied sample solution

Correct answer: b

Selected sources for this lecture

- IU Examination Office: Written Assignment Task for CSEBCSCSAS01 - Computer Science and Society.
- Meta Transparency Center. Facebook Feed AI system and approach to Feed ranking, 2025.
- European Commission. Digital Services Act information and recommender-system transparency context.
- ENISA. Threat Landscape 2024 and cybersecurity threat landscape resources.
- European Union Aviation Safety Agency (EASA). Civil drone operation categories and open/specific category guidance.
- Stahl, B. C. (2022). From computer ethics and the ethics of AI towards an ethics of digital ecosystems. AI and Ethics, 2(1), 65-77.

THANK YOU

Questions and individual topic clinic